Estimating in Building Construction

Chapter 3 Project Manual

Project Manual

- Project manual accompanies the drawings (plans)

- Often referred to as specifications
  - Project manual refers to the written document(s) that accompany the plans
  - Specification refers to the technical specifications
    - Specifications are part of project manual

CSI

- Construction Specifications Institute (CSI)
  - Sets standard format for:
    - Project manual
    - Technical specifications
  - 2004 MasterFormat™
    - 50 divisions (0 to 49)
    - See Figure 3-1 for first two levels

Contents of Project Manual

- Invitation to bid (Advertisement for bids)
  - Publicly advertized:
    - Newspapers
    - Trade journals
      - Engineering New Record (ENR)
    - Internet

Contents of Project Manual

- Invitation to bid (Advertisement for bids)
  - Describes:
    - Location and nature of work
    - Bid date and time
    - Where to obtain documents
    - Bid bond requirements
    - Start and completion dates

Contents of Project Manual

- Instruction to bidders (Information for bidders)
  - Procedures that must be followed by all bidders
    - How to submit a proposal (bid)
    - Commencement and completion dates
    - Bidders responsibilities (i.e. site visit required)
    - Award or rejection of bid

Contents of Project Manual

- Bid forms
  - Standard form that must be submitted by all bidders
  - Includes:
    - Price
    - Last date the bid may be accepted
    - Entry for all addenda received
    - Bid security
    - Liquid damages

Contents of Project Manual

- Form of owner/contractor agreement
  - Contract between owner and contractor
  - May be:
    - Standard American Institute of Architects (AIA) contract
    - Company form
    - Government agency form

Contents of Project Manual

- General conditions
  - Conditions that apply to all aspects of the project
  - May be:
    - Standard AIA general conditions
    - Company document
    - Government agency document

Contents of Project Manual

- Supplementary general conditions
  - Mechanism to the specific project
  - Modifies the general conditions

Contents of Project Manual

- Specifications (technical specifications)
  - Describes:
    - Materials (physical properties, performance requirement, Handling and storage requiremnt)
    - Construction methods
    - Workmanship and quality
  - Often follows MasterFormat™

Contents of Project Manual

- Alternates
  - Add or deduct pricing for:
    - Alternate methods of construction
    - Alternate materials

Contents of Project Manual

- Addenda
  - Modifies the contract documents after they have been released and prior to the bid
  - Important to make sure that:
    - You have seen all the addenda
    - Your subcontractors and suppliers have see all of the addenda
    - A space is provided in which the contractor must list the addenda received

Errors in the Specifications

- Errors should be brought to the attention of the architect or engineer

- Requests for clarifications of the specifications should be submitted to the engineer so that the clarification can be sent to all bidders

Thank You